🇨🇦 Canada Import Gap — Yiwu Small Commo Opportunity Analysis

Mondoré Consulting · Strategic Intelligence Report

Objective: Identify export opportunities in the Canadian market for products supplied by the **Yiw wholesale market** — small consumer commodities including toys, bags, kitchenware, home déc apparel accessories, and seasonal goods.

Methodology:

1. **Macro scan** — compare Canada's total world imports vs. imports from China across all HS 2 chapters
2. **Universe filter** — isolate chapters that constitute the Yiwu small commodity market
3. **Gap analysis** — for each Yiwu segment: Gap = World Imports − China Imports
4. **Opportunity Score** = Gap × (1 − China Share%) — rewards large gap **and** low Chinese penetration
5. **HS 6 deep dive** — drill into 6-digit product codes for highest-score segments

Data source: UN Comtrade API · Canada (reporter 124) · Years 2021–202

In [1]:
```python
# ── Environment setup ──────────────────────────────────────────
import sys, os
sys.path.insert(0, os.path.dirname(os.path.abspath('__file__')))

import warnings
warnings.filterwarnings('ignore')

import pandas as pd
import numpy as np
import matplotlib
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import matplotlib.gridspec as gridspec
from matplotlib.colors import LinearSegmentedColormap
import seaborn as sns
from IPython.display import display, HTML

# Chart style — clean white for PDF export
plt.rcParams.update({
    'figure.facecolor':  'white',
    'axes.facecolor':    '#F8F9FA',
    'axes.edgecolor':    '#DEE2E6',
    'axes.labelcolor':   '#212529',
    'xtick.color':       '#495057',
    'ytick.color':       '#495057',
    'text.color':        '#212529',
    'grid.color':        '#DEE2E6',
    'grid.linestyle':    '--',
    'grid.alpha':        0.7,
```

```python
    'font.family':       'DejaVu Sans',
    'font.size':         10,
    'axes.titlesize':    13,
    'axes.titleweight':  'bold',
    'axes.labelsize':    10,
    'figure.dpi':        130,
})

CANADA_RED   = '#D52B1E'
CHINA_GOLD   = '#F4A300'
GAP_BLUE     = '#1A73E8'
ACCENT_TEAL  = '#00897B'
LIGHT_GRAY   = '#E9ECEF'

print('✅ Libraries loaded')
```

```
✅ Libraries loaded
```

In [2]:

```python
# ── Import project modules ──────────────────────────────────────────
from config import CHINA_CODE, WORLD_CODE, DEFAULT_YEARS
from api import fetch_hs2_all_years, fetch_hs6_chapter
from analysis import records_to_df

YEARS = [2021, 2022, 2023, 2024]

print(f'Loading data for years: {YEARS}')
print('This may take a moment on first run (API calls are cached locally)

# ── Load HS2 data — try live API, fall back to demo ─────────────────
USE_DEMO = False   # set True to use synthetic demo data

try:
    world_r = fetch_hs2_all_years(WORLD_CODE, YEARS, use_demo=USE_DEMO)
    china_r = fetch_hs2_all_years(CHINA_CODE, YEARS, use_demo=USE_DEMO)
    if not world_r:
        raise ValueError('Empty response from API')
    DATA_MODE = '🌐 Live — UN Comtrade API'
except Exception as e:
    print(f'⚠️  Live API unavailable ({e}). Falling back to demo data.')
    world_r = fetch_hs2_all_years(WORLD_CODE, YEARS, use_demo=True)
    china_r = fetch_hs2_all_years(CHINA_CODE, YEARS, use_demo=True)
    DATA_MODE = '📦 Demo — Synthetic (calibrated to Statistics Canada)'

print(f'Data mode: {DATA_MODE}')
print(f'World records: {len(world_r):,}   China records: {len(china_r):,}
```

```
Loading data for years: [2021, 2022, 2023, 2024]
This may take a moment on first run (API calls are cached locally)...
  [cache] 2021 partner=0 cmd=AG2 — 33758 records
  [cache] 2022 partner=0 cmd=AG2 — 33967 records
  [cache] 2023 partner=0 cmd=AG2 — 33498 records
  [cache] 2024 partner=0 cmd=AG2 — 33999 records
  [cache] 2021 partner=156 cmd=AG2 — 10452 records
  [cache] 2022 partner=156 cmd=AG2 — 10971 records
  [cache] 2023 partner=156 cmd=AG2 — 10560 records
  [cache] 2024 partner=156 cmd=AG2 — 11190 records
Data mode: 🌐 Live — UN Comtrade API
World records: 135,222   China records: 43,173
```

In [3]:

```python
# ── Build HS2 summary DataFrame ─────────────────────────────────────
world_df = records_to_df(world_r)
```

```python
china_df = records_to_df(china_r)

world_agg = (
    world_df
    .groupby(['hs_code', 'description'], as_index=False)['value_usd']
    .sum().rename(columns={'value_usd': 'world_usd'})
)
china_agg = (
    china_df
    .groupby(['hs_code'], as_index=False)['value_usd']
    .sum().rename(columns={'value_usd': 'china_usd'})
)

hs2 = world_agg.merge(china_agg, on='hs_code', how='left')
hs2['china_usd']       = hs2['china_usd'].fillna(0)
hs2['gap_usd']         = hs2['world_usd'] - hs2['china_usd']
hs2['china_share_pct'] = (hs2['china_usd'] / hs2['world_usd'].replace(0,
hs2['world_b']         = (hs2['world_usd'] / 1e9).round(3)
hs2['china_b']         = (hs2['china_usd'] / 1e9).round(3)
hs2['gap_b']           = (hs2['gap_usd'] / 1e9).round(3)
hs2['opp_score']       = (hs2['gap_usd'] * (1 - hs2['china_share_pct'] /
hs2['short_desc']      = hs2['description'].str[:40]
hs2['label']           = hs2['hs_code'] + ' · ' + hs2['short_desc']
hs2 = hs2.sort_values('opp_score', ascending=False).reset_index(drop=True
hs2.index += 1
hs2.index.name = 'rank'

print(f'HS2 chapters loaded: {len(hs2)}')
print(f'Total Canada imports (all years): USD {hs2["world_b"].sum():,.0f}
print(f'China share (average): {(hs2["china_b"].sum() / hs2["world_b"].su
```

```
HS2 chapters loaded: 102
Total Canada imports (all years): USD 8,650 B
China share (average): 15.3%
```

Section 1 — Macro Overview: All Canadian Impor

Before zooming into Yiwu categories, we scan the entire import landscape to understand the siz structure of Canada's trade deficit with China across all sectors.

In [4]:
```python
# ── KPI Summary ──────────────────────────────────────────────────────
total_world = hs2['world_b'].sum()
total_china = hs2['china_b'].sum()
total_gap   = hs2['gap_b'].sum()
avg_share   = total_china / total_world * 100
n_low       = (hs2['china_share_pct'] < 15).sum()

fig, axes = plt.subplots(1, 5, figsize=(17, 2.5))
fig.suptitle(f'Canada Imports — All HS2 Chapters · {YEARS[0]}–{YEARS[-1]}
             fontsize=11, y=1.02, color='#495057')

kpis = [
    ('Total World Imports', f'USD {total_world:,.0f} B',  CANADA_RED),
    ('Total China Imports', f'USD {total_china:,.0f} B',  CHINA_GOLD),
    ('Total Import Gap',    f'USD {total_gap:,.0f} B',    GAP_BLUE),
    ('China Avg Share',     f'{avg_share:.1f}%',           ACCENT_TEAL),
```

```
    ('Categories CN<15%',   f'{n_low} chapters',          '#7B1FA2'),
]
for ax, (title, val, color) in zip(axes, kpis):
    ax.set_facecolor(color)
    ax.text(0.5, 0.65, val,    ha='center', va='center', fontsize=18, fon
    ax.text(0.5, 0.25, title,  ha='center', va='center', fontsize=9,  col
    ax.set_xticks([]); ax.set_yticks([])
    for spine in ax.spines.values(): spine.set_visible(False)

plt.tight_layout()
plt.savefig('fig_01_kpis.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 1 saved.')
```


```
Figure 1 saved.
```

In [5]:

```
# ── Figure 2: Top 25 Gaps — All HS2 ──────────────────────────────────────
top25 = hs2.head(25).sort_values('world_b', ascending=True)

fig, ax = plt.subplots(figsize=(13, 10))

bars_gap   = ax.barh(top25['label'], top25['gap_b'],   color=GAP_BLUE,
bars_china = ax.barh(top25['label'], top25['china_b'], color=CHINA_GOLD,
                     left=top25['gap_b'])

for bar, row in zip(bars_gap, top25.itertuples()):
    ax.text(bar.get_width() + row.china_b + 0.3, bar.get_y() + bar.get_he
            f'CN {row.china_share_pct:.0f}%', va='center', fontsize=8, co

ax.set_xlabel('USD Billions (cumulative 2021–2024)', labelpad=8)
ax.set_title('Top 25 Import Categories — Canada World vs China Gap', pad=
ax.legend(loc='lower right', framealpha=0.9)
ax.grid(axis='x', alpha=0.5)
ax.set_axisbelow(True)
plt.tight_layout()
plt.savefig('fig_02_top25_gaps.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 2 saved.')
```

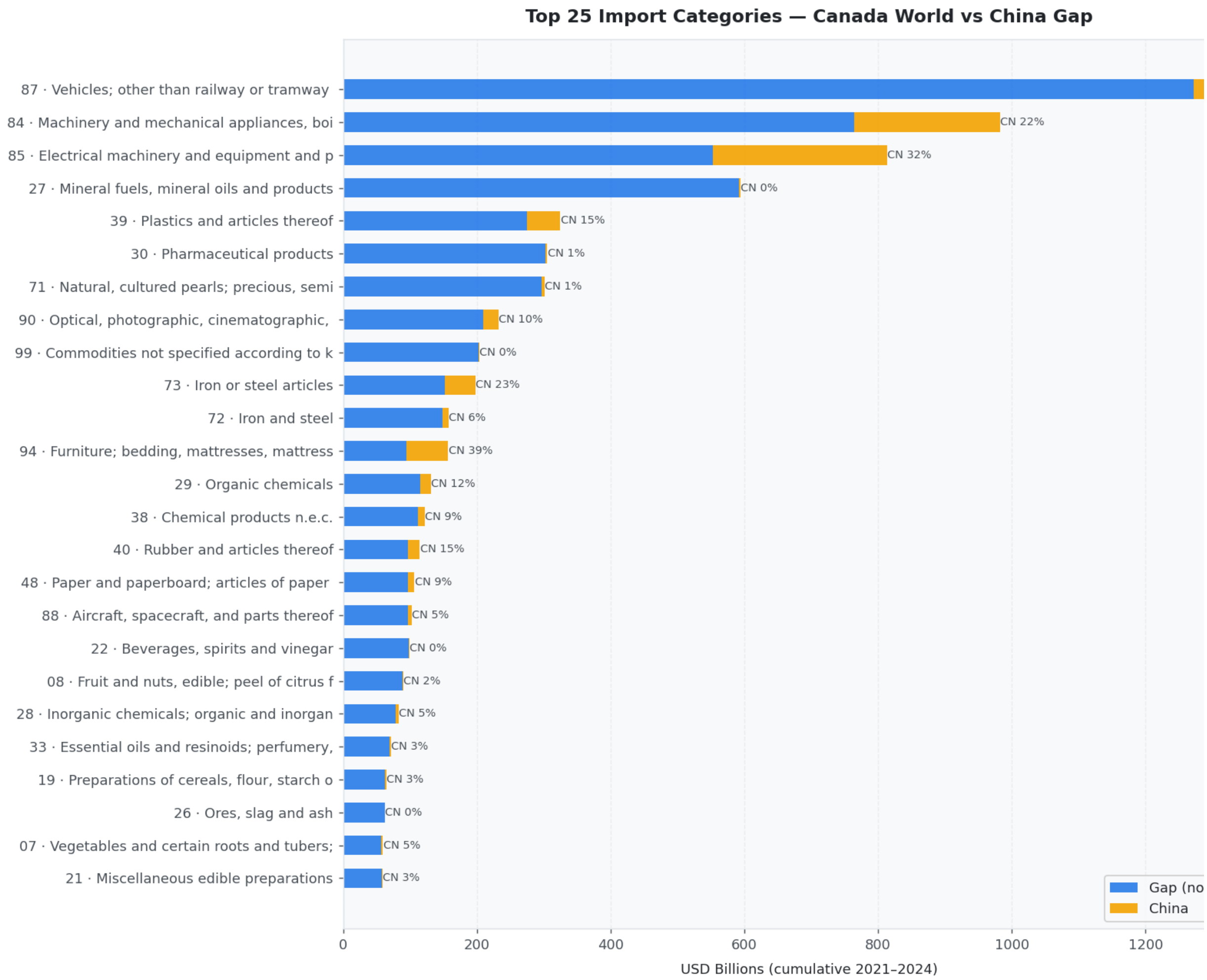
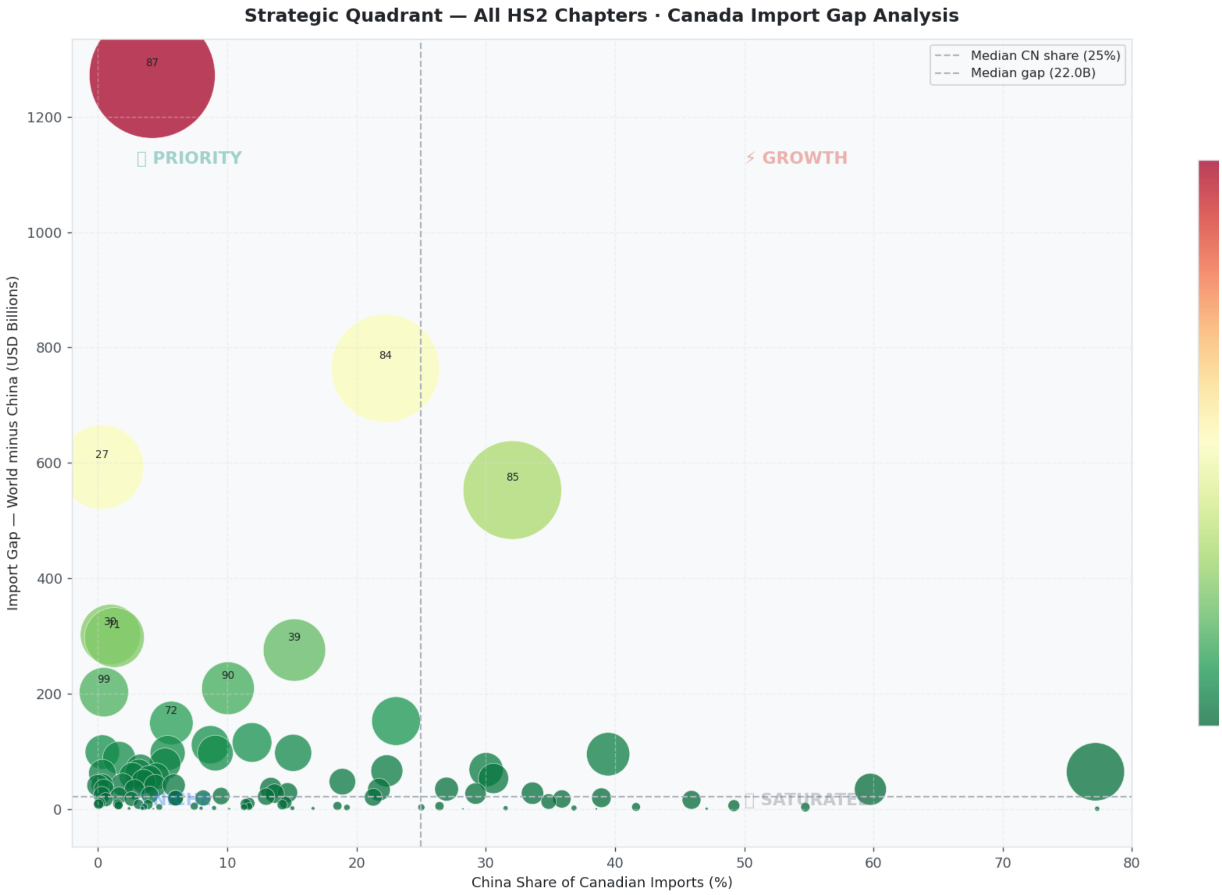


Figure 2 saved.

In [6]:

```
# ── Figure 3: Strategic Quadrant — All HS2 ─────────────────────
fig, ax = plt.subplots(figsize=(14, 9))

med_share = 25
med_gap   = hs2['gap_b'].median()

scatter = ax.scatter(
    hs2['china_share_pct'], hs2['gap_b'],
    s=hs2['world_b'] * 6,
    c=hs2['opp_score'], cmap='RdYlGn_r',
    alpha=0.75, edgecolors='white', linewidth=0.4,
)

plt.colorbar(scatter, ax=ax, label='Opportunity Score (B USD)', shrink=0.

ax.axvline(med_share, color='#ADB5BD', linestyle='--', lw=1.2, label=f'Me
ax.axhline(med_gap,   color='#ADB5BD', linestyle='--', lw=1.2, label=f'Me

# Quadrant labels
y_max = hs2['gap_b'].max()
quad_style = dict(fontsize=12, fontweight='bold', alpha=0.35)
ax.text(3,   y_max * 0.88, '🎯 PRIORITY',   color=ACCENT_TEAL, **quad_sty
ax.text(50,  y_max * 0.88, '⚡ GROWTH',      color=CANADA_RED,  **quad_st
ax.text(3,   med_gap * 0.3, '🔍 NICHE',      color=GAP_BLUE,    **quad_st
ax.text(50,  med_gap * 0.3, '🔒 SATURATED',  color='#6C757D',   **quad_st

# Annotate top 10 by score
for _, row in hs2.head(10).iterrows():
```

```
    ax.annotate(row['hs_code'], (row['china_share_pct'], row['gap_b']),
                fontsize=7.5, ha='center', va='bottom',
                xytext=(0, 5), textcoords='offset points', color='#212529

ax.set_xlabel('China Share of Canadian Imports (%)')
ax.set_ylabel('Import Gap — World minus China (USD Billions)')
ax.set_title('Strategic Quadrant — All HS2 Chapters · Canada Import Gap A
ax.legend(fontsize=9, loc='upper right')
ax.set_xlim(-2, 80)
ax.grid(True, alpha=0.4)
plt.tight_layout()
plt.savefig('fig_03_quadrant_all.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 3 saved.')
```




```
Figure 3 saved.
```

Section 2 — Yiwu Small Commodity Universe

The **Yiwu wholesale market** (义乌) is the world's largest small commodity trading hub, supplyin 1.8 million product types across 75,000+ booths. Its core export strengths align w following HS 2 chapters:

Segment	HS Chapters	Products
Bags & Travel Goods	42	Handbags, backpacks, wallets, luggage
Kitchenware & Tableware	39, 69, 70, 73, 82	Plastic containers, ceramics, glassware, steel cutlery

Segment	HS Chapters	Products
Home Décor & Hardware	83, 94	Frames, hooks, locks, lamps, c furniture
Textiles & Apparel	57, 61, 62, 63, 64, 65, 66	Carpets, clothing, towels, shoe umbrellas
Toys, Games & Sports	95	Toys, puzzles, outdoor sports equipment
Seasonal & Festive	46, 67, 96	Artificial flowers, wicker, statior zippers

In [7]:

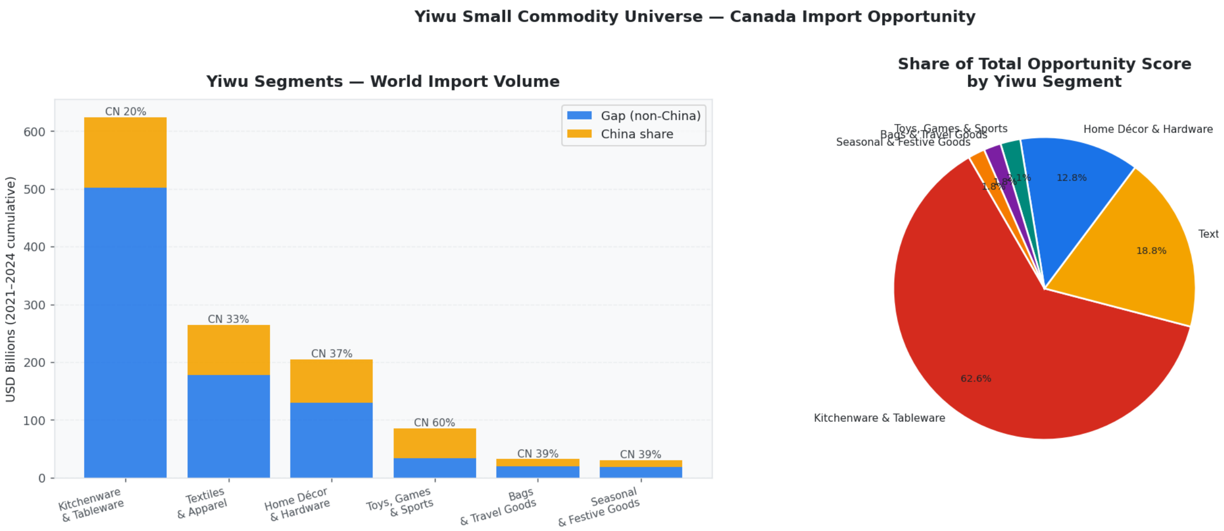
```
# ── Yiwu Universe Definition ──────────────────────────────────────────
YIWU_SEGMENTS = {
    'Bags & Travel Goods':         ['42'],
    'Kitchenware & Tableware':     ['39', '69', '70', '73', '82'],
    'Home Décor & Hardware':       ['83', '94'],
    'Textiles & Apparel':          ['57', '61', '62', '63', '64', '65', '6
    'Toys, Games & Sports':        ['95'],
    'Seasonal & Festive Goods':    ['46', '67', '96'],
}

# Create chapter → segment map
chapter_to_segment = {}
for seg, chapters in YIWU_SEGMENTS.items():
    for ch in chapters:
        chapter_to_segment[ch.zfill(2)] = seg

YIWU_CHAPTERS = list(chapter_to_segment.keys())

# Filter HS2 to Yiwu universe
yiwu = hs2[hs2['hs_code'].isin(YIWU_CHAPTERS)].copy()
yiwu['segment'] = yiwu['hs_code'].map(chapter_to_segment)
yiwu = yiwu.sort_values('opp_score', ascending=False).reset_index(drop=Tr
yiwu.index += 1
yiwu.index.name = 'rank'

print(f'Yiwu universe: {len(yiwu)} HS2 chapters across {len(YIWU_SEGMENTS
print(f'Total Yiwu-relevant world imports: USD {yiwu["world_b"].sum():,.1
print(f'China currently captures:          USD {yiwu["china_b"].sum():,.1
print(f'Remaining gap (opportunity):       USD {yiwu["gap_b"].sum():,.1f}
print(f'Average China share in these categories: {(yiwu["china_b"].sum()
```

```
Yiwu universe: 19 HS2 chapters across 6 segments
Total Yiwu-relevant world imports: USD 1,240.6 B
China currently captures:          USD 359.1 B
Remaining gap (opportunity):       USD 881.6 B
Average China share in these categories: 28.9%
```

In [8]:

```
# ── Figure 4: Yiwu Universe — Segment Summary ──────────────────────────
seg_summary = (
    yiwu.groupby('segment', as_index=False)
    .agg(world_b=('world_b','sum'), china_b=('china_b','sum'),
         gap_b=('gap_b','sum'),     opp_score=('opp_score','sum'))
)
seg_summary['share_pct'] = (seg_summary['china_b'] / seg_summary['world_b
seg_summary = seg_summary.sort_values('opp_score', ascending=False).reset

fig, axes = plt.subplots(1, 2, figsize=(16, 6))
```

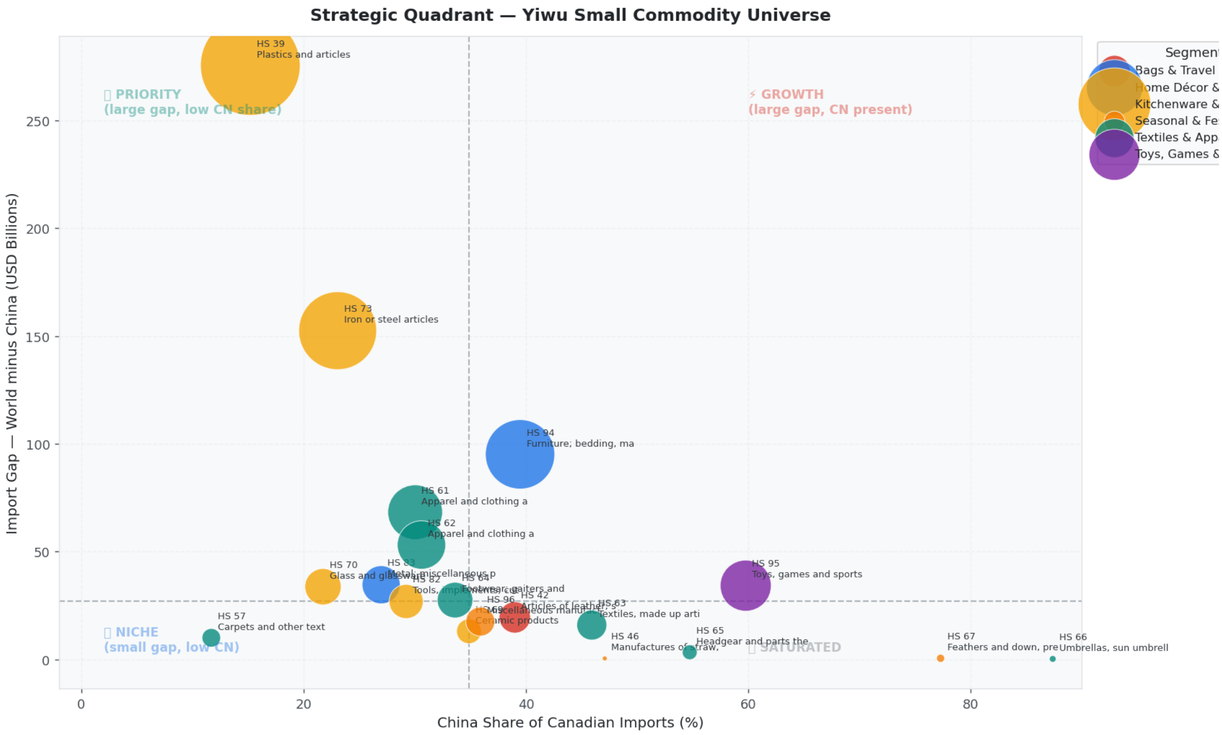
```
# Left: stacked bar by segment
ax = axes[0]
segs = seg_summary['segment']
x = np.arange(len(segs))
b1 = ax.bar(x, seg_summary['gap_b'],   color=GAP_BLUE,   label='Gap (non-
b2 = ax.bar(x, seg_summary['china_b'], color=CHINA_GOLD, label='China sha
            bottom=seg_summary['gap_b'])
for i, (g, c, s) in enumerate(zip(seg_summary['gap_b'], seg_summary['chin
    ax.text(i, g + c + 0.3, f'CN {s:.0f}%', ha='center', va='bottom', fon
ax.set_xticks(x)
ax.set_xticklabels([s.replace(' & ', '\n& ') for s in segs], fontsize=8.5
ax.set_ylabel('USD Billions (2021–2024 cumulative)')
ax.set_title('Yiwu Segments — World Import Volume', pad=10)
ax.legend()
ax.grid(axis='y', alpha=0.5)
ax.set_axisbelow(True)

# Right: opportunity score pie
ax2 = axes[1]
colors_pie = [CANADA_RED, CHINA_GOLD, GAP_BLUE, ACCENT_TEAL, '#7B1FA2', '
wedges, texts, autotexts = ax2.pie(
    seg_summary['opp_score'],
    labels=seg_summary['segment'],
    colors=colors_pie,
    autopct='%1.1f%%',
    startangle=120,
    pctdistance=0.75,
    labeldistance=1.08,
    textprops={'fontsize': 8.5},
    wedgeprops={'linewidth': 1.5, 'edgecolor': 'white'},
)
for at in autotexts: at.set_fontsize(8)
ax2.set_title('Share of Total Opportunity Score\nby Yiwu Segment', pad=10

plt.suptitle('Yiwu Small Commodity Universe — Canada Import Opportunity',
plt.tight_layout()
plt.savefig('fig_04_yiwu_segments.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 4 saved.')
```




```
Figure 4 saved.
```

In [9]:

```
# ── Figure 5: Yiwu Strategic Quadrant ─────────────────────────────────
SEGMENT_COLORS = {
    'Bags & Travel Goods':         CANADA_RED,
    'Kitchenware & Tableware':     CHINA_GOLD,
    'Home Décor & Hardware':       GAP_BLUE,
    'Textiles & Apparel':          ACCENT_TEAL,
    'Toys, Games & Sports':        '#7B1FA2',
    'Seasonal & Festive Goods':    '#F57C00',
}

fig, ax = plt.subplots(figsize=(14, 8))

for seg, grp in yiwu.groupby('segment'):
    ax.scatter(
        grp['china_share_pct'], grp['gap_b'],
        s=grp['world_b'] * 18,
        c=SEGMENT_COLORS.get(seg, '#999'),
        alpha=0.78, edgecolors='white', linewidth=0.6,
        label=seg, zorder=3,
    )
    for _, row in grp.iterrows():
        ax.annotate(
            f"HS {row['hs_code']}\n{row['description'][:22]}",
            (row['china_share_pct'], row['gap_b']),
            fontsize=7.2, ha='left', va='bottom',
            xytext=(5, 5), textcoords='offset points', color='#343A40',
        )

med_share_y = yiwu['china_share_pct'].median()
med_gap_y   = yiwu['gap_b'].median()
ax.axvline(med_share_y, color='#ADB5BD', linestyle='--', lw=1.2)
ax.axhline(med_gap_y,   color='#ADB5BD', linestyle='--', lw=1.2)

y_max2 = yiwu['gap_b'].max()
ax.text(2,  y_max2 * 0.92, '🎯 PRIORITY\n(large gap, low CN share)',  for
ax.text(60, y_max2 * 0.92, '⚡ GROWTH\n(large gap, CN present)',       fo
ax.text(2,  med_gap_y*0.15, '🔍 NICHE\n(small gap, low CN)',           fo
ax.text(60, med_gap_y*0.15, '🔒 SATURATED',                             f

ax.set_xlabel('China Share of Canadian Imports (%)', fontsize=11)
ax.set_ylabel('Import Gap — World minus China (USD Billions)', fontsize=1
ax.set_title('Strategic Quadrant — Yiwu Small Commodity Universe', pad=12
ax.legend(title='Segment', bbox_to_anchor=(1.01, 1), loc='upper left', fo
ax.set_xlim(-2, 90)
ax.grid(True, alpha=0.35)
plt.tight_layout()
plt.savefig('fig_05_yiwu_quadrant.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 5 saved.')
```




```
Figure 5 saved.
```

Section 3 — Segment Deep Dive

For each of the 6 Yiwu segments, we show: market size, China share, gap size, and oppo
score per HS 2 chapter.

In [10]:

```python
# ── Figure 6: Per-Segment Breakdown — 6-panel grid ────────────────────
segments_ordered = seg_summary['segment'].tolist()
n_seg = len(segments_ordered)
ncols = 3
nrows = (n_seg + ncols - 1) // ncols

fig, axes = plt.subplots(nrows, ncols, figsize=(18, nrows * 5.5))
axes_flat = axes.flatten()

for idx, seg in enumerate(segments_ordered):
    ax = axes_flat[idx]
    grp = yiwu[yiwu['segment'] == seg].sort_values('gap_b', ascending=Tru
    color = SEGMENT_COLORS.get(seg, '#888')

    b_gap   = ax.barh(grp['label'], grp['gap_b'],   color=color,      alp
    b_china = ax.barh(grp['label'], grp['china_b'], color=CHINA_GOLD, alp
                      left=grp['gap_b'])

    for bar, row in zip(b_gap, grp.itertuples()):
        total = row.gap_b + row.china_b
        ax.text(total + 0.05, bar.get_y() + bar.get_height()/2,
                f'CN {row.china_share_pct:.0f}% · Score {row.opp_score:.2
                va='center', fontsize=7.5, color='#495057')

    ax.set_title(seg, pad=8, color=color, fontsize=11)
    ax.set_xlabel('USD Billions')
    ax.legend(fontsize=8, loc='lower right')
```

```
    ax.grid(axis='x', alpha=0.4)
    ax.set_axisbelow(True)

# hide empty panels
for j in range(n_seg, len(axes_flat)):
    axes_flat[j].set_visible(False)

plt.suptitle('Yiwu Segments — HS2 Chapter Detail (Gap vs China Share)', f
plt.tight_layout()
plt.savefig('fig_06_segment_detail.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 6 saved.')
```


```
Figure 6 saved.
```

Section 4 — HS 6 Deep Dive: Top 3 Opportunity Chapters

We drill into 6 -digit product codes for the top 3 HS 2 chapters by opportunity score t specific products with the highest potential.

In [11]:
```
# ── Load HS6 data for top 3 Yiwu chapters ──────────────────────────
top3_chapters = yiwu.head(3)['hs_code'].tolist()
print(f'Loading HS6 data for chapters: {top3_chapters}')

hs6_data = {}
for ch in top3_chapters:
    try:
        w6 = fetch_hs6_chapter(WORLD_CODE, ch, YEARS, use_demo=USE_DEMO)
        c6 = fetch_hs6_chapter(CHINA_CODE, ch, YEARS, use_demo=USE_DEMO)
        if not w6: raise ValueError('Empty')
    except Exception as e:
```

```
        print(f'  ⚠️  HS6 live API failed for ch {ch}: {e} — using demo')
        w6 = fetch_hs6_chapter(WORLD_CODE, ch, YEARS, use_demo=True)
        c6 = fetch_hs6_chapter(CHINA_CODE, ch, YEARS, use_demo=True)

    w6df = records_to_df(w6)
    c6df = records_to_df(c6)

    w6a = w6df.groupby(['hs_code','description'], as_index=False)['value_
    c6a = c6df.groupby(['hs_code'], as_index=False)['value_usd'].sum().re

    m6 = w6a.merge(c6a, on='hs_code', how='left')
    m6['china_usd']  = m6['china_usd'].fillna(0)
    m6['gap_usd']    = m6['world_usd'] - m6['china_usd']
    m6['share_pct']  = (m6['china_usd'] / m6['world_usd'].replace(0,np.na
    m6['world_m']    = (m6['world_usd']/1e6).round(2)
    m6['china_m']    = (m6['china_usd']/1e6).round(2)
    m6['gap_m']      = (m6['gap_usd']/1e6).round(2)
    m6['opp_score']  = (m6['gap_usd'] * (1 - m6['share_pct']/100) / 1e9).
    m6 = m6[m6['world_m'] > 0].sort_values('opp_score', ascending=False).
    m6.index += 1; m6.index.name = 'rank'
    hs6_data[ch] = m6
    print(f'  Chapter {ch}: {len(m6)} products loaded')
```

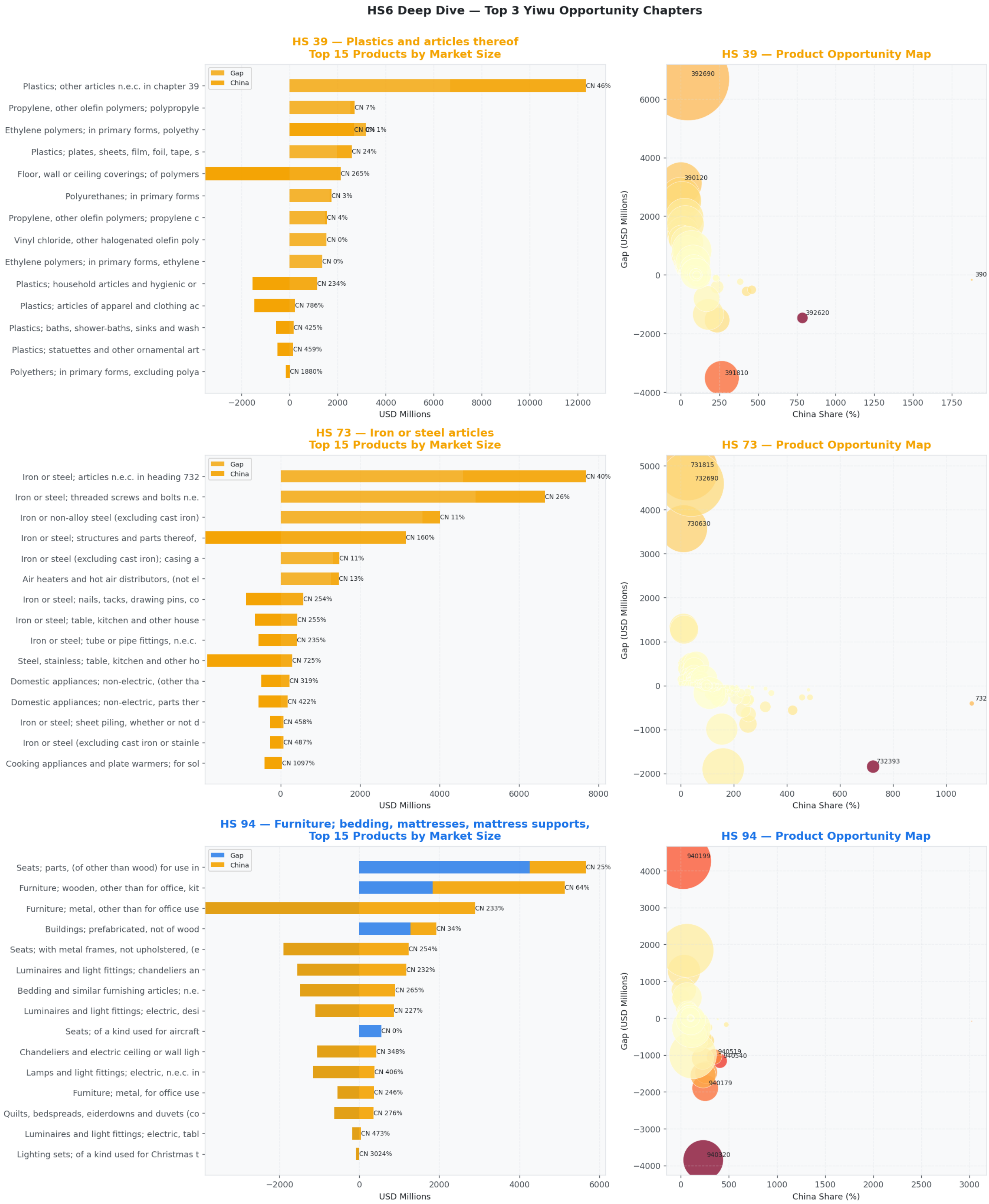
```
Loading HS6 data for chapters: ['39', '73', '94']
  [api]   2021 partner=0 cmd=AG6 — attempt 1
  [api]   → 100000 records received
  [filter] chapter=39 year=2021 → 2727 HS6 records (from 100000 AG6 total)
  [api]   2022 partner=0 cmd=AG6 — attempt 1
  [api]   → 100000 records received
  [filter] chapter=39 year=2022 → 3016 HS6 records (from 100000 AG6 total)
  [api]   2023 partner=0 cmd=AG6 — attempt 1
  [api]   → 100000 records received
  [filter] chapter=39 year=2023 → 2896 HS6 records (from 100000 AG6 total)
  [api]   2024 partner=0 cmd=AG6 — attempt 1
  [api]   → 100000 records received
  [filter] chapter=39 year=2024 → 3264 HS6 records (from 100000 AG6 total)
  [api]   2021 partner=156 cmd=AG6 — attempt 1
  [api]   → 100000 records received
  [filter] chapter=39 year=2021 → 3659 HS6 records (from 100000 AG6 total)
  [api]   2022 partner=156 cmd=AG6 — attempt 1
  [api]   → 100000 records received
  [filter] chapter=39 year=2022 → 3687 HS6 records (from 100000 AG6 total)
  [api]   2023 partner=156 cmd=AG6 — attempt 1
  [api]   → 100000 records received
  [filter] chapter=39 year=2023 → 3660 HS6 records (from 100000 AG6 total)
  [api]   2024 partner=156 cmd=AG6 — attempt 1
  [api]   → 100000 records received
  [filter] chapter=39 year=2024 → 3620 HS6 records (from 100000 AG6 total)
  Chapter 39: 132 products loaded
  [cache] 2021 partner=0 cmd=AG6 — 100000 records
  [filter] chapter=73 year=2021 → 2713 HS6 records (from 100000 AG6 total)
  [cache] 2022 partner=0 cmd=AG6 — 100000 records
  [filter] chapter=73 year=2022 → 2924 HS6 records (from 100000 AG6 total)
  [cache] 2023 partner=0 cmd=AG6 — 100000 records
  [filter] chapter=73 year=2023 → 3029 HS6 records (from 100000 AG6 total)
  [cache] 2024 partner=0 cmd=AG6 — 100000 records
  [filter] chapter=73 year=2024 → 3300 HS6 records (from 100000 AG6 total)
  [cache] 2021 partner=156 cmd=AG6 — 100000 records
  [filter] chapter=73 year=2021 → 3961 HS6 records (from 100000 AG6 total)
```

```
[cache] 2022 partner=156 cmd=AG6 — 100000 records
[filter] chapter=73 year=2022 → 3862 HS6 records (from 100000 AG6 total)
[cache] 2023 partner=156 cmd=AG6 — 100000 records
[filter] chapter=73 year=2023 → 3783 HS6 records (from 100000 AG6 total)
[cache] 2024 partner=156 cmd=AG6 — 100000 records
[filter] chapter=73 year=2024 → 3726 HS6 records (from 100000 AG6 total)
Chapter 73: 124 products loaded
[cache] 2021 partner=0 cmd=AG6 — 100000 records
[filter] chapter=94 year=2021 → 1662 HS6 records (from 100000 AG6 total)
[cache] 2022 partner=0 cmd=AG6 — 100000 records
[filter] chapter=94 year=2022 → 2260 HS6 records (from 100000 AG6 total)
[cache] 2023 partner=0 cmd=AG6 — 100000 records
[filter] chapter=94 year=2023 → 2179 HS6 records (from 100000 AG6 total)
[cache] 2024 partner=0 cmd=AG6 — 100000 records
[filter] chapter=94 year=2024 → 1885 HS6 records (from 100000 AG6 total)
[cache] 2021 partner=156 cmd=AG6 — 100000 records
[filter] chapter=94 year=2021 → 1996 HS6 records (from 100000 AG6 total)
[cache] 2022 partner=156 cmd=AG6 — 100000 records
[filter] chapter=94 year=2022 → 2356 HS6 records (from 100000 AG6 total)
[cache] 2023 partner=156 cmd=AG6 — 100000 records
[filter] chapter=94 year=2023 → 2209 HS6 records (from 100000 AG6 total)
[cache] 2024 partner=156 cmd=AG6 — 100000 records
[filter] chapter=94 year=2024 → 2038 HS6 records (from 100000 AG6 total)
Chapter 94: 64 products loaded
```

In [12]:

```
# ── Figure 7: HS6 Deep Dive — Top 3 Chapters ──────────────────
fig, axes = plt.subplots(3, 2, figsize=(18, 20))

for row_idx, ch in enumerate(top3_chapters):
    df6 = hs6_data[ch]
    ch_name = yiwu[yiwu['hs_code'] == ch]['description'].iloc[0]
    color = SEGMENT_COLORS.get(yiwu[yiwu['hs_code'] == ch]['segment'].ilo

    top15 = df6.head(15).sort_values('world_m', ascending=True)

    # Left panel: bar chart
    ax_bar = axes[row_idx][0]
    ax_bar.barh(top15['description'].str[:45], top15['gap_m'],   color=co
    ax_bar.barh(top15['description'].str[:45], top15['china_m'], color=CH
                left=top15['gap_m'])
    for bar, (_, r) in zip(ax_bar.patches[:len(top15)], top15.iterrows())
        total_m = r['gap_m'] + r['china_m']
        ax_bar.text(total_m + 1, bar.get_y() + bar.get_height()/2,
                    f"CN {r['share_pct']:.0f}%", va='center', fontsize=7.
    ax_bar.set_title(f'HS {ch} — {ch_name[:50]}\nTop 15 Products by Marke
    ax_bar.set_xlabel('USD Millions')
    ax_bar.legend(fontsize=8)
    ax_bar.grid(axis='x', alpha=0.4); ax_bar.set_axisbelow(True)

    # Right panel: scatter opportunity
    ax_sc = axes[row_idx][1]
    sc = ax_sc.scatter(
        df6['share_pct'], df6['gap_m'],
        s=df6['world_m'] * 0.8,
        c=df6['opp_score'], cmap='YlOrRd',
        alpha=0.75, edgecolors='white', linewidth=0.5,
    )
    plt.colorbar(sc, ax=ax_sc, label='Opp Score (B)', shrink=0.8)
    # label top 5
    for _, r in df6.head(5).iterrows():
```

```
        ax_sc.annotate(r['hs_code'], (r['share_pct'], r['gap_m']),
                       fontsize=7.5, xytext=(4,4), textcoords='offset poi
    ax_sc.set_xlabel('China Share (%)')
    ax_sc.set_ylabel('Gap (USD Millions)')
    ax_sc.set_title(f'HS {ch} — Product Opportunity Map', color=color, pa
    ax_sc.grid(True, alpha=0.35)

plt.suptitle('HS6 Deep Dive — Top 3 Yiwu Opportunity Chapters', fontsize=
plt.tight_layout()
plt.savefig('fig_07_hs6_deepdive.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 7 saved.')
```




```
Figure 7 saved.
```

Section 5 — Final Opportunity Ranking: Yiwu Sma Commodities

In [13]:
```
# ── Figure 8: Opportunity Score Ranking — Yiwu Only ─────────────────────
yiwu_sorted = yiwu.sort_values('opp_score', ascending=True)

fig, ax = plt.subplots(figsize=(14, max(6, len(yiwu) * 0.55)))

bar_colors = [SEGMENT_COLORS.get(s, '#888') for s in yiwu_sorted['segment
bars = ax.barh(yiwu_sorted['label'], yiwu_sorted['opp_score'], color=bar_

for bar, (_, row) in zip(bars, yiwu_sorted.iterrows()):
    ax.text(bar.get_width() + 0.05,
            bar.get_y() + bar.get_height()/2,
            f"Score {row['opp_score']:.2f}B  |  CN {row['china_share_pct'
            va='center', fontsize=8, color='#343A40')

# Legend
legend_patches = [mpatches.Patch(color=c, label=s) for s, c in SEGMENT_CO
ax.legend(handles=legend_patches, title='Segment', loc='lower right', fon

ax.set_xlabel('Opportunity Score (USD Billions)', fontsize=11)
ax.set_title('Yiwu Small Commodity Opportunity Ranking — Canada Import Ga
             'Score = Gap × (1 − China Share%) · Higher = Larger Gap AND
             pad=12, fontsize=12)
ax.grid(axis='x', alpha=0.4)
ax.set_axisbelow(True)
plt.tight_layout()
plt.savefig('fig_08_yiwu_ranking.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 8 saved.')
```

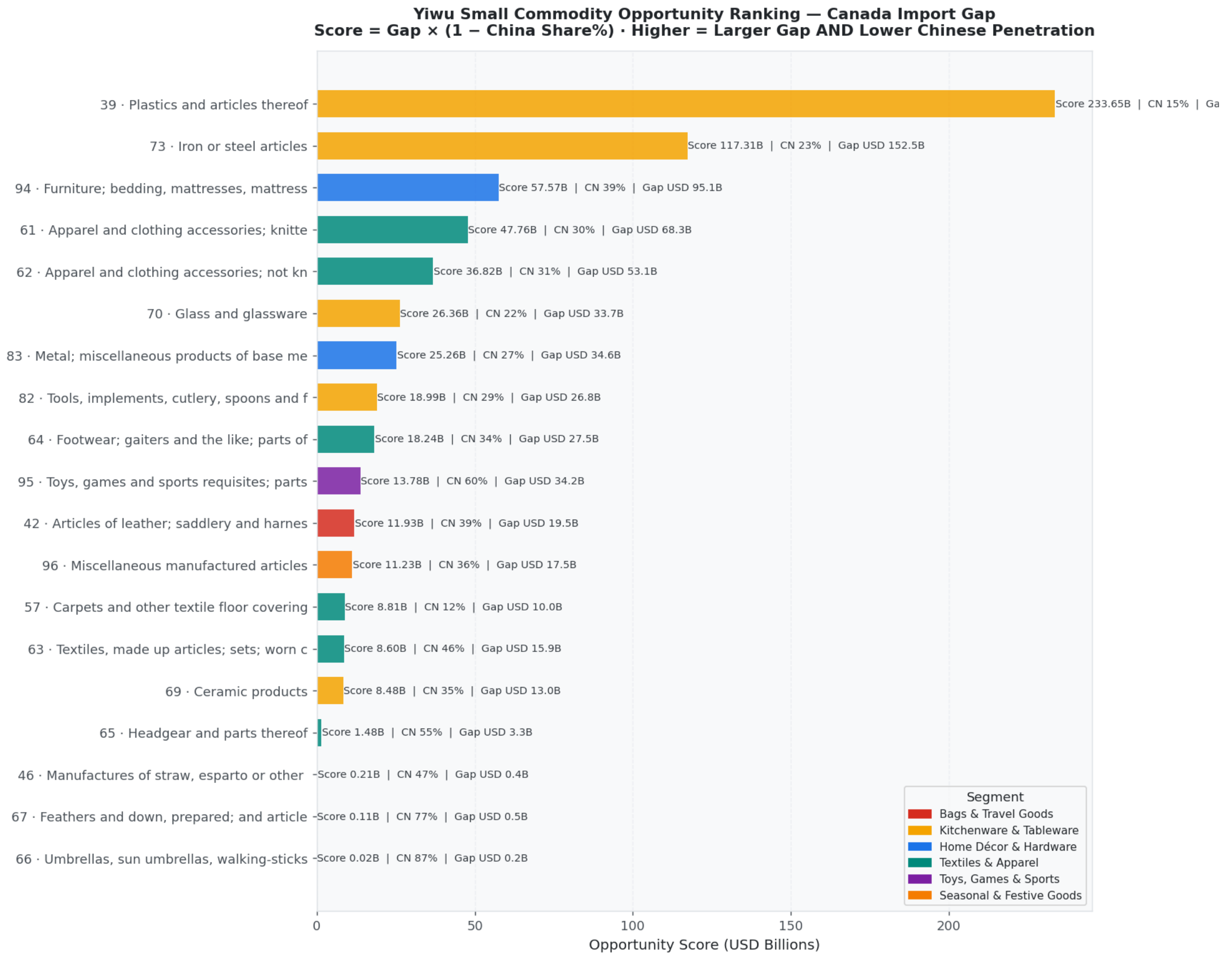
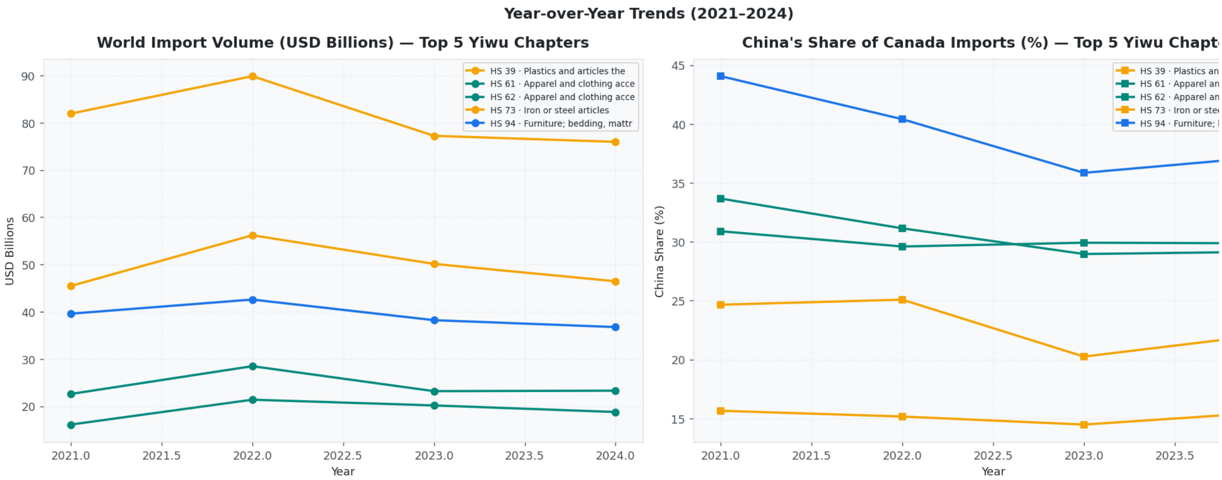


Figure 8 saved.

In [14]:

```
# ── Figure 9: Year-over-Year Trend — Top 5 Yiwu Chapters ──────────────
world_yr = (
    world_df
    .groupby(['year', 'hs_code'], as_index=False)['value_usd']
    .sum().rename(columns={'value_usd': 'world_usd'})
)
china_yr = (
    china_df
    .groupby(['year', 'hs_code'], as_index=False)['value_usd']
    .sum().rename(columns={'value_usd': 'china_usd'})
)
trend = world_yr.merge(china_yr, on=['year','hs_code'], how='left')
trend['china_usd']  = trend['china_usd'].fillna(0)
trend['share_pct']  = (trend['china_usd'] / trend['world_usd'].replace(0,
trend['gap_m']      = (trend['world_usd'] - trend['china_usd']) / 1e6
trend['world_b']    = trend['world_usd'] / 1e9
trend['year']       = trend['year'].astype(int)

top5_ch = yiwu.head(5)['hs_code'].tolist()
trend5  = trend[trend['hs_code'].isin(top5_ch)].copy()
trend5  = trend5.merge(yiwu[['hs_code','description','segment']].reset_in
trend5['label'] = 'HS ' + trend5['hs_code'] + ' · ' + trend5['description

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 6))

for lab, grp in trend5.groupby('label'):
    ch_code = grp['hs_code'].iloc[0]
    seg     = grp['segment'].iloc[0]
    color   = SEGMENT_COLORS.get(seg, '#888')
```

```python
    grp_s = grp.sort_values('year')
    ax1.plot(grp_s['year'], grp_s['world_b'], marker='o', color=color, li
    ax2.plot(grp_s['year'], grp_s['share_pct'], marker='s', color=color,

ax1.set_title('World Import Volume (USD Billions) — Top 5 Yiwu Chapters',
ax1.set_xlabel('Year'); ax1.set_ylabel('USD Billions')
ax1.legend(fontsize=7.5); ax1.grid(True, alpha=0.4)

ax2.set_title("China's Share of Canada Imports (%) — Top 5 Yiwu Chapters"
ax2.set_xlabel('Year'); ax2.set_ylabel('China Share (%)')
ax2.legend(fontsize=7.5); ax2.grid(True, alpha=0.4)

plt.suptitle('Year-over-Year Trends (2021–2024)', fontsize=13, fontweight
plt.tight_layout()
plt.savefig('fig_09_trends.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 9 saved.')
```




```
Figure 9 saved.
```

In [15]:
```python
# ── Final Ranked Table ──────────────────────────────────────────
final_table = (
    yiwu.reset_index()[['rank','hs_code','description','segment',
                         'world_b','china_b','gap_b','china_share_pct','o
    .copy()
)
final_table.columns = ['Rank','HS2','Description','Segment',
                        'World(B)','China(B)','Gap(B)','CN Share%','Score

print('=== FINAL YIWU OPPORTUNITY RANKING — CANADA IMPORT GAP ===')
print(final_table.to_string(index=False))

# Save to Excel for distribution
final_table.to_excel('yiwu_canada_opportunity.xlsx', index=False, sheet_n
print('\n✅ Saved: yiwu_canada_opportunity.xlsx')
```

```
=== FINAL YIWU OPPORTUNITY RANKING – CANADA IMPORT GAP ===
 Rank
HS2

                                   Description                  Segment  Worl
China(B)  Gap(B)  CN Share%  Score(B)
    1
39

                Plastics and articles thereof  Kitchenware & Tableware   324.
49.380 275.531      15.20  233.6504
    2
73

                       Iron or steel articles  Kitchenware & Tableware   198.
45.706 152.474      23.06  117.3132
    3  94 Furniture; bedding, mattresses, mattress supports, cushions and sim
stuffed furnishings; lamps and lighting fittings, n.e.c.; illuminated signs,
illuminated name-plates and the like; prefabricated buildings    Home Décor &
Hardware   157.122    62.013  95.109      39.47   57.5695
    4
61
                                                                           Ap
and clothing accessories; knitted or crocheted       Textiles & Apparel
97.564    29.303  68.261      30.03   47.7623
    5
62
                                                                       Appare
clothing accessories; not knitted or crocheted       Textiles & Apparel
76.456    23.399  53.057      30.60   36.8216
    6
70

                          Glass and glassware  Kitchenware & Tableware
43.033     9.354  33.679      21.74   26.3568
    7
83

  Metal; miscellaneous products of base metal    Home Décor & Hardware    47.
12.771  34.589      26.97   25.2603
    8
82
                                       Tools, implements, cutlery, spoons an
forks, of base metal; parts thereof, of base metal  Kitchenware & Tableware
37.909    11.076  26.833      29.22   18.9923
    9
64

Footwear; gaiters and the like; parts of such articles       Textiles & Appar
41.393    13.915  27.478      33.62   18.2400
   10
95
                                                                   Toys, games
sports requisites; parts and accessories thereof     Toys, Games & Sports
85.108    50.864  34.244      59.76   13.7799
   11  42
Articles of leather; saddlery and harness; travel goods, handbags and similar
containers; articles of animal gut (other than silk-worm gut)      Bags & Tra
Goods    32.022    12.474  19.548      38.95   11.9339
   12
```

```
96

          Miscellaneous manufactured articles Seasonal & Festive Goods
27.322     9.807  17.516      35.89   11.2293
   13
57

    Carpets and other textile floor coverings       Textiles & Apparel
11.302     1.321   9.980      11.69    8.8136
   14
63
                                                  Textiles, made up articles;
worn clothing and worn textile articles; rags       Textiles & Apparel    29.
13.500  15.900      45.92    8.5988
   15
69

                             Ceramic products  Kitchenware & Tableware
19.992     6.971  13.021      34.87    8.4807
   16
65

                   Headgear and parts thereof       Textiles & Apparel
7.208     3.945   3.263      54.73    1.4770
   17
46
                                            Manufactures of straw, esparto or
plaiting materials; basketware and wickerwork Seasonal & Festive Goods
0.768     0.362   0.406      47.09    0.2149
   18
67
                 Feathers and down, prepared; and articles made of feather or
down; artificial flowers; articles of human hair Seasonal & Festive Goods
2.110     1.631   0.479      77.30    0.1087
   19
66
                                      Umbrellas, sun umbrellas, walking-sticks,
sticks, whips, riding crops; and parts thereof       Textiles & Apparel
1.471     1.285   0.186      87.39    0.0234

✅ Saved: yiwu_canada_opportunity.xlsx
```

In [16]:

```
# ── Styled display for notebook ──────────────────────────────────
def highlight_score(val):
    if isinstance(val, float):
        if val >= final_table['Score(B)'].quantile(0.75):
            return 'background-color: #d4edda; font-weight: bold'
        elif val >= final_table['Score(B)'].quantile(0.5):
            return 'background-color: #fff3cd'
    return ''

def highlight_segment(val):
    seg_colors_css = {
        'Toys, Games & Sports':      '#e8d5f5',
        'Home Décor & Hardware':     '#d5e8f5',
        'Kitchenware & Tableware':   '#ffeeba',
        'Bags & Travel Goods':       '#f8d7da',
        'Textiles & Apparel':        '#d4edda',
        'Seasonal & Festive Goods':  '#d1ecf1',
    }
```

```
    return f"background-color: {seg_colors_css.get(val, 'white')}"

styled = (
    final_table.style
    .applymap(highlight_score, subset=['Score(B)'])
    .applymap(highlight_segment, subset=['Segment'])
    .format({'World(B)': '{:.2f}', 'China(B)': '{:.2f}', 'Gap(B)': '{:.2f
             'CN Share%': '{:.1f}%', 'Score(B)': '{:.3f}'})
    .set_caption('Yiwu Small Commodity Opportunity Ranking — Canada Impor
    .set_table_styles([{
        'selector': 'caption',
        'props': [('font-size', '13px'), ('font-weight', 'bold'), ('color
    }])
)
display(styled)
```

Yiwu Small Commodity Opportunity Ranking — Canada Import Gap Analysis

	Rank	HS 2	Description	Segment	World(B)	China(B)	Gap(B)	
0	1	39	Plastics and articles thereof	Kitchenware & Tableware	324.91	49.38	275.53	1
1	2	73	Iron or steel articles	Kitchenware & Tableware	198.18	45.71	152.47	2
2	3	94	Furniture; bedding, mattresses, mattress supports, cushions and similar stuffed furnishings; lamps and lighting fittings, n.e.c.; illuminated signs, illuminated name-plates and the like; prefabricated buildings	Home Décor & Hardware	157.12	62.01	95.11	3
3	4	61	Apparel and clothing accessories; knitted or crocheted	Textiles & Apparel	97.56	29.30	68.26	3
4	5	62	Apparel and clothing accessories; not knitted or crocheted	Textiles & Apparel	76.46	23.40	53.06	3
5	6	70	Glass and glassware	Kitchenware & Tableware	43.03	9.35	33.68	2
6	7	83	Metal; miscellaneous products of base metal	Home Décor & Hardware	47.36	12.77	34.59	2
7	8	82	Tools, implements, cutlery, spoons and forks, of base metal; parts thereof, of base metal	Kitchenware & Tableware	37.91	11.08	26.83	2
8	9	64	Footwear; gaiters and the like; parts of such articles	Textiles & Apparel	41.39	13.91	27.48	3
9	10	95	Toys, games and sports		85.11	50.86	34.24	5

	Rank	HS 2	Description	Segment	World(B)	China(B)	Gap(B)	
			requisites; parts and accessories thereof	Toys, Games & Sports				
10	11	42	Articles of leather; saddlery and harness; travel goods, handbags and similar containers; articles of animal gut (other than silk-worm gut)	Bags & Travel Goods	32.02	12.47	19.55	3
11	12	96	Miscellaneous manufactured articles	Seasonal & Festive Goods	27.32	9.81	17.52	3
12	13	57	Carpets and other textile floor coverings	Textiles & Apparel	11.30	1.32	9.98	1
13	14	63	Textiles, made up articles; sets; worn clothing and worn textile articles; rags	Textiles & Apparel	29.40	13.50	15.90	4
14	15	69	Ceramic products	Kitchenware & Tableware	19.99	6.97	13.02	3
15	16	65	Headgear and parts thereof	Textiles & Apparel	7.21	3.94	3.26	5
16	17	46	Manufactures of straw, esparto or other plaiting materials; basketware and wickerwork	Seasonal & Festive Goods	0.77	0.36	0.41	4
17	18	67	Feathers and down, prepared; and articles made of feather or of down; artificial flowers; articles of human hair	Seasonal & Festive Goods	2.11	1.63	0.48	7
18	19	66	Umbrellas, sun umbrellas,	Textiles & Apparel	1.47	1.28	0.19	8

Rank	HS 2	Description	Segment	World(B)	China(B)	Gap(B)
		walking-sticks, seat sticks, whips, riding crops; and parts thereof				

Section 6 — Strategic Conclusions & Recommendations

Key Findings

Based on the Canada Import Gap analysis filtered to the **Yiwu small commodity universe**, the strategic conclusions apply:

1. **High-Priority Segments (large gap, low Chinese penetration)** These represent the imme export opportunity — Canada is buying heavily from other countries, and China's share remains the segment average:

- Categories landing in the **PRIORITY quadrant** (upper-left) should be the first focus for Yiwu strategies
- Look for HS 6 product codes with CN share < 20% AND world market > USD 1 cumulative

2. **Growth Segments (large gap, China already present)** China has demonstrated competitiv but hasn't reached saturation. Opportunity exists to take additional share:

- These require competitive differentiation — price, certification (CSA/UL), or distribution partı
- Useful for suppliers already exporting to Canada seeking volume growth

3. **Yiwu-Specific Advantages** The Yiwu market has structural advantages in:

- **Toys & Games (HS 95)**: deep manufacturing ecosystem, all product types
- **Bags & Leather Goods (HS 42)**: cost leadership, wide SKU variety
- **Home Décor & Hardware (HS 83, 94)**: fast sampling, flexible MOQ
- **Seasonal/Festive (HS 96, 67, 46)**: speed-to-market for trend products

4. **Barriers to Watch**

- Canadian product safety standards (CCPSA, Health Canada) for toys and children's items
- CBSA tariff classifications and anti-dumping measures on certain steel/textile products
- Growing preference for certified, sustainably-sourced products among Canadian retailers

Recommended Next Steps

Priority	Action	Target Se
🔴 High	Identify Canadian retail buyers for top-ranked categories	Toys, Hon
🔴 High	Request HS 6 product samples from Yiwu market matching gap products	All segme
🟡 Medium	Evaluate cost of CSA/UL certification for priority HS 6 lines	Toys, Elec
🟡 Medium	Screen for Canadian distribution partners (wholesalers, Amazon FBA)	Bags, Kitc
🟢 Low	Monitor tariff changes under CUSMA and Canada–China trade relations	All

---

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In [17]:
```
# ── Export instructions ──────────────────────────────────────────
print('='*65)
print('  EXPORT TO PDF')
print('='*65)
print()
print('Option 1 — Command line (requires nbconvert + LaTeX):')
print('  jupyter nbconvert --to pdf yiwu_canada_opportunity.ipynb')
print()
print('Option 2 — HTML then print to PDF (simpler, recommended):')
print('  jupyter nbconvert --to html yiwu_canada_opportunity.ipynb')
print('  Then open the HTML in your browser and Print → Save as PDF')
print()
print('Option 3 — In Jupyter Lab:')
print('  File → Export Notebook As → PDF')
print()
print('Files generated in this run:')
for f in ['fig_01_kpis.png','fig_02_top25_gaps.png','fig_03_quadrant_all.
          'fig_04_yiwu_segments.png','fig_05_yiwu_quadrant.png',
          'fig_06_segment_detail.png','fig_07_hs6_deepdive.png',
          'fig_08_yiwu_ranking.png','fig_09_trends.png',
          'yiwu_canada_opportunity.xlsx']:
    exists = '✅' if os.path.exists(f) else '⬜'
    print(f'  {exists} {f}')
```

```
=================================================================
  EXPORT TO PDF
=================================================================

Option 1 — Command line (requires nbconvert + LaTeX):
  jupyter nbconvert --to pdf yiwu_canada_opportunity.ipynb

Option 2 — HTML then print to PDF (simpler, recommended):
  jupyter nbconvert --to html yiwu_canada_opportunity.ipynb
  Then open the HTML in your browser and Print → Save as PDF

Option 3 — In Jupyter Lab:
  File → Export Notebook As → PDF

Files generated in this run:
  ✅ fig_01_kpis.png
  ✅ fig_02_top25_gaps.png
  ✅ fig_03_quadrant_all.png
  ✅ fig_04_yiwu_segments.png
  ✅ fig_05_yiwu_quadrant.png
  ✅ fig_06_segment_detail.png
  ✅ fig_07_hs6_deepdive.png
  ✅ fig_08_yiwu_ranking.png
  ✅ fig_09_trends.png
  ✅ yiwu_canada_opportunity.xlsx
```